Medical Diagnosis Using ANN

A PROJECT REPORT
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Submitted by

Prabhav Kansal
(202410116100143)
Priyanshu Kansal
(202410116100155)
Sagar Gupta
(202410116100174)

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Mr. Apoorv Jain
Assistant Professor

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CERTIFICATE

Certified that **Prabhav Kansal (202410116100143), Priyanshu Kansal (202410116100155), Sagar Gupta (202410116100174)** has/ have carried out the project work having **“Medical Diagnosis using ANN ” (AI Project, AI101B)** for Master of Computer Applications from Dr. A.P.J. Abdul Kalam Technical University (AKTU) (formerly UPTU), Lucknow under my supervision. The project report embodies original work, and studies are carried out by the student himself/herself and the contents of the project report do not form the basis for the award of any other degree to the candidate or to anybody else from this or any other University/Institution.

This is certify that the above statement made by the candidate is correct to the best of my knowledge.

Date:

Mr. Apoorv Jain

Assistant Professor

Department of Computer Applications
KIET Group of Institutions, Ghaziabad
An Autonomous Institutions

Dr. Akash Rajak

Dean

Department of Computer Applications
KIET Group of Institutions, Ghaziabad
An Autonomous Institutions

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Prabhav Kansal

Priyanshu Kansal

Sagar Gupta

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1. INTRODUCTION

1.1 Overview

Breast cancer is one of the most prevalent and deadly diseases among women worldwide. Early and accurate diagnosis is essential for effective treatment and improved survival rates. Traditional diagnostic methods, such as mammography, biopsy, and clinical examination, rely heavily on expert interpretation and are susceptible to errors like false positives or false negatives. These limitations can lead to delayed treatment, misdiagnosis, and increased patient anxiety.

Artificial Intelligence (AI), particularly **Artificial Neural Networks (ANNs)**, provides a powerful solution to enhance diagnostic accuracy and efficiency. ANNs are computational models inspired by the human brain and are capable of learning complex patterns from large datasets. This project focuses on developing an intelligent ANN-based model that predicts whether a breast tumor is benign (non-cancerous) or malignant (cancerous) using labeled medical data.

The model is trained on features extracted from patient records, such as cell radius, texture, perimeter, and smoothness—typically obtained from fine needle aspiration tests. By analyzing these features, the ANN learns the underlying differences between benign and malignant tumors. Once trained, the model can deliver rapid, consistent, and highly accurate predictions.

Additionally, we integrate a real-time user input interface that allows new patient data to be evaluated using the trained ANN model. This demonstrates a practical use case for deploying such predictive systems in telehealth platforms, healthcare apps, or decision-support tools for doctors.

This AI-driven system supports healthcare professionals by reducing diagnostic time, minimizing errors, and providing a reliable second opinion. It is particularly useful in remote or under-resourced areas lacking specialized medical staff. Additionally, it can be integrated into clinical decision support systems or mobile health applications.
In conclusion, this project demonstrates the potential of ANN in transforming breast cancer diagnosis. It not only enhances early detection but also paves the way for smarter, data-driven healthcare solutions that can save lives and improve treatment outcomes.

1.2 Importance of Predicting Heart Disease

Breast cancer remains one of the leading causes of cancer-related deaths in women worldwide. Its high mortality rate is primarily due to late diagnosis, where treatment becomes less effective. Early detection significantly improves survival rates, making timely and accurate diagnosis essential. However, traditional diagnostic techniques such as mammograms and biopsies rely on human interpretation, which can be time-consuming and prone to errors like false positives or negatives.

The use of Artificial Neural Networks (ANNs) in this project introduces a powerful and intelligent approach to improving breast cancer diagnosis. ANNs can learn complex patterns in medical data, allowing them to detect subtle indicators of malignancy that may be missed by conventional methods. By analyzing patient features like cell radius, texture, compactness, and smoothness, an ANN model can deliver accurate predictions on whether a tumor is benign or malignant.

One of the key advantages of this system is its ability to reduce human error and diagnostic variability. The ANN provides consistent results regardless of fatigue, experience, or subjective judgment. This enhances the reliability of diagnoses, particularly in regions where access to expert oncologists is limited.

Moreover, ANN-based diagnosis is faster and more scalable. It can process thousands of patient records in a fraction of the time it would take a human specialist, making it ideal for use in busy hospitals or during mass screening campaigns. It also lowers healthcare costs by minimizing unnecessary procedures and treatments caused by misdiagnosis.

Overall, this project has the potential to transform medical diagnostics by offering a data-driven, automated, and highly accurate system for breast cancer prediction. It supports doctors in making better decisions and empowers healthcare systems to deliver timely care, ultimately helping to save lives.

1.3 Scope of the Project

The scope of this project encompasses the development, evaluation, and potential deployment of an **Artificial Neural Network (ANN)-based system** for predicting breast cancer. The aim is to build a reliable diagnostic tool that supports medical professionals in identifying whether a breast tumor is benign or malignant, based on patient data. This project's scope can be broadly categorized into four key areas:

1. Technical Scope

- **Dataset Utilization**: Use of publicly available medical datasets such as the Wisconsin Breast Cancer Diagnostic (WBCD) dataset containing features like radius, texture, perimeter, and smoothness of cell nuclei.
- **Data Preprocessing**: Cleaning, normalization, and transformation of data to ensure high-quality input for the neural network.
- **Model Development**: Designing and training an ANN using appropriate architectures, activation functions (ReLU, sigmoid), and optimization techniques (like Adam).
- **Performance Evaluation**: Using accuracy, precision, recall, F1-score, and ROC-AUC metrics to evaluate model performance.
- **Deployment Readiness**: Designing the model to be scalable and integrable into real-world healthcare platforms.

2. Clinical and Healthcare Scope

- **Decision Support**: Assisting doctors with an AI-based second opinion to improve diagnostic accuracy and confidence.
- **Early Detection Tool**: Offering rapid analysis to detect cancer in its early stages, leading to timely treatment.
- **Rural and Remote Access**: Providing diagnostic support in areas lacking specialized oncologists or medical infrastructure.
- **Telemedicine Integration**: Potential to be integrated into mobile health apps or telemedicine platforms for real-time diagnostics.

3. Educational and Research Scope

- **Student and Research Use**: Useful for academic research, AI learning, and practical applications in medical data science.
- **Foundation for Advanced Models**: Scope to extend the project using deep learning, CNNs (for image-based diagnostics), and hybrid AI models.

2. LITERATURE REVIEW

Introduction

Breast cancer diagnosis using machine learning and artificial intelligence has been an active area of research for several years. Many studies have demonstrated the effectiveness of Artificial Neural Networks (ANNs) in identifying patterns in complex medical datasets, enabling accurate classification of breast tumors as benign or malignant.

1. Historical Context and Early Methods

Initial research into automated breast cancer diagnosis began with the use of statistical methods, such as logistic regression and decision trees. Although useful, these techniques often struggled to handle the non-linear relationships and high-dimensional nature of medical data. Researchers soon began exploring Artificial Neural Networks, which proved more adept at learning complex patterns.

In a foundational study, Street et al. (1993) utilized the Wisconsin Breast Cancer Dataset (WBCD) and demonstrated that ANNs could achieve over 95% accuracy in classifying tumors. This was a significant milestone, showing that machine learning could assist in clinical diagnosis.

2. Comparative Studies with Other ML Models

Various machine learning algorithms, such as Support Vector Machines (SVM), K-Nearest Neighbors (KNN), and Random Forests, have been compared with ANN models for breast cancer prediction. In many cases, ANNs showed superior or comparable performance when properly tuned and trained.

For instance, Patil and Kumar (2015) compared ANN and SVM models and found that ANNs performed better in sensitivity (true positive rate), making them more suitable for medical applications where missing a positive case could be fatal.

3. ANN Architectures in Medical Diagnosis

Recent research has explored deep neural networks (DNNs), which include multiple hidden layers to enhance learning. These models are capable of detecting intricate features within data and improving prediction accuracy. However, for tabular data like the WBCD, shallow ANN models (1–2 hidden layers) are often sufficient and computationally efficient.

4. Role of Data Quality and Preprocessing

Several studies, including those by Sasikala and Suresh (2018), emphasized the importance of data preprocessing, such as normalization and feature selection, in improving model performance. Clean, balanced, and noise-free datasets significantly boost the accuracy and robustness of ANN predictions.

2.1 Artificial Neural Networks in Healthcare Prediction

ANNs mimic the biological neural networks of the human brain and are capable of learning complex patterns through training on labeled data. Early studies in the 1990s began exploring ANNs for cancer diagnosis, demonstrating their potential to improve classification accuracy compared to conventional statistical methods.

- Street et al. (1993) were among the pioneers using the Wisconsin Breast Cancer Dataset (WBCD). They showed that a well-configured ANN achieved an accuracy of over 95% in classifying tumors as benign or malignant.
- ANNs' capacity to model non-linear relationships makes them superior to simpler linear classifiers for medical data, which often have complex underlying patterns.

2.2 Ensemble Models and Hybrid Techniques

In recent years, researchers have explored **ensemble models** and **hybrid techniques** to enhance the performance of breast cancer prediction systems beyond what single models like standalone Artificial Neural Networks (ANNs) can achieve. These approaches combine multiple classifiers or integrate different algorithms to leverage their unique strengths and improve accuracy, robustness, and generalization.

Ensemble Models
Ensemble learning involves combining several base models to form a stronger predictive model. Popular ensemble methods include:

- **Bagging (Bootstrap Aggregating):** Multiple models are trained on different subsets of the training data. For example, Random Forest, an ensemble of decision trees, has been successfully applied to breast cancer diagnosis with high accuracy.
- **Boosting:** Sequentially trains models, where each new model focuses on correcting the errors of the previous ones. Algorithms like AdaBoost and Gradient Boosting Machines have been used alongside ANNs to improve classification.

Hybrid Techniques
Hybrid techniques combine different algorithms or data processing methods to capitalize on their complementary advantages. Common hybrid approaches in breast cancer prediction include:

- **ANN + Fuzzy Logic:** Incorporating fuzzy logic with ANN helps handle uncertainty and imprecise data common in medical diagnosis. This hybrid model can better mimic expert decision-making and improve classification accuracy.
- **ANN + Genetic Algorithms (GA):** Genetic Algorithms optimize ANN parameters such as weights, learning rate, or network architecture, enhancing the network's performance and convergence speed.

2.3 Real-Time and Mobile Health Applications

Real-time applications involve systems that can process data instantly or within a very short time frame, enabling immediate or near-immediate diagnostic feedback. In breast cancer prediction, real-time ANN models can analyze patient data such as imaging results or biopsy features during clinical visits, helping doctors make fast and informed decisions.

- **Clinical Decision Support Systems (CDSS):** ANNs integrated into hospital software allow doctors to input patient data and receive instant predictions on tumor malignancy, reducing diagnosis time and enhancing treatment planning.

- **Radiology Assistance:** Real-time ANN-based tools can assist radiologists by flagging suspicious lesions in mammograms or ultrasound scans during the screening process, improving detection rates and reducing human errors.

- **Telemedicine:** Especially useful in remote areas, real-time ANN systems enable doctors to upload patient data for immediate analysis, facilitating timely interventions.

Mobile Health (mHealth) Applications

- **Portable Diagnostic Tools:** Mobile apps embedded with ANN models can analyze uploaded images or patient data and provide preliminary breast cancer risk assessments anytime and anywhere.

- **Self-Screening and Awareness:** These apps empower individuals to perform regular self-checks by answering symptom questionnaires or capturing images, with AI models offering risk predictions and suggesting when to seek professional care.

2.4 Datasets and Evaluation Metrics

1. **Wisconsin Breast Cancer Diagnostic (WBCD) Dataset**
    - One of the most widely used datasets in breast cancer prediction research.
    - Contains 569 instances with 30 numeric features extracted from digitized images of fine needle aspirate (FNA) of breast masses.
    - Features include radius, texture, perimeter, area, smoothness, compactness, concavity, symmetry, and fractal dimension.
    - Labels classify tumors as **benign** (357 cases) or **malignant** (212 cases).

2. **Wisconsin Breast Cancer Original (WBC) Dataset**
    - Earlier version of WBCD, contains fewer features and samples but still valuable for comparative studies.

2.5 Relevance to Present Study

The present study aims to develop an effective Artificial Neural Network (ANN) model for accurate and early prediction of breast cancer, addressing critical challenges in current diagnostic methods. The relevance of existing research, datasets, and evaluation techniques to this study is outlined below:

Building on Established Foundations

- The majority of breast cancer prediction research relies on benchmark datasets such as the Wisconsin Breast Cancer Diagnostic (WBCD) dataset. This study leverages the same high-quality dataset, ensuring that the developed ANN model can be compared with prior work and validated against a standard set of features and labels.
- By adopting well-known evaluation metrics (accuracy, precision, recall, F1-score, ROC-AUC), the study ensures rigorous performance assessment and aligns with best practices observed in the literature.

Addressing Limitations in Traditional Diagnosis

- Traditional diagnostic techniques often suffer from subjective interpretation and diagnostic delays. The present study aims to automate this process by employing ANN's ability to learn complex, non-linear patterns from patient data, thereby reducing reliance on human judgment and potentially lowering false positive and false negative rates.
- Prior studies indicate that while ANN models are effective, challenges remain in optimizing network architecture and training processes. This study explores these areas in detail to improve prediction accuracy and model robustness.

Conclusion

The literature demonstrates that Artificial Neural Networks are highly effective tools for breast cancer prediction, capable of learning from historical patient data and producing accurate results. While other algorithms also perform well, ANNs offer a flexible and powerful framework for classification tasks in medical domains. Continued research is focusing on improving interpretability, ensuring fairness, and integrating these tools into real-time clinical systems.

3. SYSTEM ANALYSIS AND REQUIREMENTS

Breast cancer diagnosis typically involves manual analysis of clinical data and imaging by medical professionals, which can be time-consuming and prone to errors due to subjective interpretations. Early and accurate detection is critical but challenging due to complex and subtle patterns in the data.

3.1 Functional Requirements

Functional requirements define what the system should do — its core operations, behavior, and functions. For this heart disease prediction system, the following functions are mandatory:

3.1.1 Data Input and Loading

- The system must allow loading of structured datasets in CSV format.
- The user must be able to manually input patient data for prediction.

3.1.2 Data Preprocessing

- The system must normalize numerical input features using StandardScaler.
- The dataset must be split into training and testing subsets.

3.1.3 Model Training

- The system must train a Random Forest classifier and an Artificial Neural Network (ANN) using the processed training data.

3.1.4 Prediction

- The system must accurately predict whether the patient is at risk (high or low) of death due to heart failure.
- The system must return a probability score between 0 and 1 for ANN output.

3.1.5 Model Evaluation

- The system must display performance metrics: accuracy, precision, recall, F1-score, and confusion matrix.
- Training graphs (accuracy/loss vs epochs) must be visualized.

3.1.6 User Prediction Interface

- The system must allow users to enter medical parameters via command-line or GUI.
- The system must return an interpretive risk assessment after prediction.

3.2 Non-Functional Requirements

Non-functional requirements define system quality, usability, performance, and operational constraints.

3.2.1 Performance

- The system must complete training of both models within a few seconds on standard hardware (≤ 10s for 299 samples).
- Prediction latency for a new user input should not exceed 1 second.

3.2.2 Usability

- The interface must be user-friendly for non-technical users.
- Input prompts should be clearly labeled and guided.

3.2.3 Reliability

- The model should provide consistent predictions across multiple runs using the same seed/random state.
- Must gracefully handle invalid inputs.

3.2.4 Maintainability

- Code must be modular and documented to allow easy updates or enhancements.

3.2.5 Scalability

- The system should be able to scale to larger datasets with minimal code changes.
- ANN architecture should be flexible for deeper or wider networks.

3.2.6 Portability

- Must run on Windows, Linux, or macOS with a Python interpreter installed.
- Should not depend on platform-specific components.

3.2.7 Security (optional/extendable)

- User input and medical predictions should be kept confidential in future deployments (e.g., HTTPS web apps).

3.3 Hardware Requirements

To run the models and visualization components smoothly, the following **minimum** and **recommended** hardware configurations are advised:

Component	Minimum Requirement	Recommended Configuration
Processor	Intel Core i3 / AMD Ryzen 3	Intel Core i5 or above
RAM	4 GB	8–16 GB
Storage	256 GB HDD or SSD	512 GB SSD
GPU	Integrated Graphics	NVIDIA GTX 1050 or higher (optional)
Operating System	Windows 10 / Linux / macOS	Windows 11 / Ubuntu 22.04 / macOS M1
Display	13” Monitor	15” or larger Full HD Display
Internet (Optional)	Required for dataset download	Stable broadband recommended

The system is lightweight and does not require high-end GPUs for small-scale datasets, making it ideal for students, researchers, and small clinics.

4. METHODOLOGY

4.1 Problem Identification & Objective Setting

The primary goal is to develop a reliable and automated diagnostic model using Artificial Neural Networks (ANN) to classify breast tumors as benign or malignant. This will assist medical professionals in making quick, data-driven decisions and improve early detection rates.

Objectives:

- Utilize real-world datasets to train an ANN model.
- Achieve high accuracy, sensitivity, and specificity.
- Reduce false positives and false negatives.
- Develop a system suitable for real-time or mobile integration.

4.1.1 Data Acquisition

Data acquisition is the foundational step in developing an effective AI-based prediction system. For this project, we use the Wisconsin Breast Cancer Diagnostic (WBCD) dataset, which is one of the most trusted and widely used datasets in medical machine learning research.

Dataset Source

- **UCI Machine Learning Repository**
- Developed by: Dr. William H. Wolberg, University of Wisconsin Hospitals

Dataset Details

- **Total Samples**: 569
- **Features**: 30 numerical attributes derived from fine needle aspirate (FNA) of breast tissue.
- **Label**: Binary classification
  - **Malignant (1)**
  - **Benign (0)**

Data Loading & Preparation

- **Tools Used**: Python, Pandas, NumPy
- **Steps:**
  - Load dataset from CSV
  - Encode diagnosis labels (M = 1, B = 0)
  - Normalize feature values using Min-Max or Standard Scaling
  - Split into **training** and **testing** sets (typically 80:20)

Why This Dataset?

- Clean, well-structured, and pre-labeled
- Balanced class distribution
- Publicly available and ethically approved for research
- Ideal for supervised learning and ANN-based classification

4.1.2 Data Preprocessing

Data preprocessing is a crucial step to prepare the dataset for training the ANN model. It ensures data quality, consistency, and optimal model performance.

Steps Involved:

1. **Data Loading**

   Imported the Wisconsin Breast Cancer Diagnostic (WBCD) dataset using Pandas.

2. **Label Encoding**

   Converted diagnosis labels:
   - Malignant (M) → 1
   - Benign (B) → 0

3. **Removing Irrelevant Columns**

   Dropped ID and unnamed columns not useful for prediction.

4. **Missing Value Handling**

   Checked for missing or null values and removed or imputed them.

5. **Feature Scaling**

   Applied StandardScaler to normalize the feature range for better ANN performance.

6. **Train-Test Split**

   Divided the dataset:
   - 80% for training
   - 20% for testing

4.1.3 Random Forest Classifier (Traditional Machine Learning)

Random Forest is a **robust ensemble method** known for its performance on tabular data.

Model Design – Artificial Neural Network (ANN)

Architecture

- **Input Layer**: 30 neurons (one for each feature)
- **Hidden Layers**: 1–2 hidden layers (e.g., 16 and 8 neurons) with ReLU activation
- **Output Layer**: 1 neuron with Sigmoid activation (outputs probability between 0 and 1)

Hyperparameters

- **Optimizer**: Adam
- **Loss Function**: Binary Crossentropy
- **Epochs:** 100–200
- **Batch Size**: 32
- **Learning Rate**: 0.001 (can be tuned)

Tools & Frameworks

- TensorFlow/Keras for ANN modeling and training

4.1.4 Artificial Neural Network (ANN)

An Artificial Neural Network (ANN) is a computational model inspired by the structure and functioning of the human brain. It consists of layers of interconnected "neurons" that can earn complex patterns from data. ANNs are particularly effective in medical diagnosis tasks due to their ability to model nonlinear relationships and recognize patterns in large datasets.

Role of ANN in Breast Cancer Prediction

In this project, ANN is used to classify whether a breast tumor is benign (non-cancerous) or malignant (cancerous) based on 30 input features (e.g., radius, texture, smoothness). The ANN learns from historical, labeled data and makes predictions on unseen patient data.

ANN Architecture

1. **Input Layer**

   30 neurons (one for each feature in the dataset)
   Accepts the normalized data from the preprocessing step.

2. **Hidden Layers**

   1 to 3 hidden layers (depending on tuning)
   Each layer contains 16 to 32 neurons.
   Activation Function: ReLU (Rectified Linear Unit) for non-linear transformation.

3. **Output Layer**

   1 neuron with Sigmoid activation function.
   Outputs a probability value between 0 and 1:
   - 0.5 → Malignant
   - ≤ 0.5 → Benign

Model Summary

Component	Configuration
Input Neurons	30
Hidden Layers	1–3 layers with 16–32 neurons each
Output Neuron	1 (binary classification)
Activation	ReLU (hidden), Sigmoid (output)
Loss Function	Binary Cross-Entropy
Optimizer	Adam
Epochs	50–100 (tuned for best accuracy)
Batch Size	16 or 32
Accuracy	~95–98% (on test data)

4.1.5 Model Training and Evaluation

Training Setup

- **Epochs**: 100–200 (or until convergence)
- **Batch Size:** 32
- **Early Stopping:** Monitors validation loss to prevent overfitting
- **Dropout Regularization:** Randomly disables neurons during training to avoid overfitting

Validation

- Use validation data to tune hyperparameters (learning rate, layers, dropout rate).
- Observe training/validation accuracy and loss curves to monitor learning behavior.

Metrics

- **Accuracy**: Overall correctness of predictions
- **Precision**: Correctness of positive predictions (malignant)
- **Recall (Sensitivity)**: Ability to detect actual positives
- **F1-Score**: Harmonic mean of precision and recall
- **Specificity**: Ability to correctly detect negatives (benign)
- **ROC Curve & AUC**: Evaluates model across various thresholds

Confusion Matrix
Provides insight into TP, FP, FN, and TN, helping understand classification quality in medical context.

4.1.5 Visualization

1. Training & Validation Accuracy and Loss Curves

- Plot accuracy and loss over epochs for both training and validation sets.
- Helps visualize model learning progress and detect overfitting
.

Example plot elements:

- X-axis: Epochs
- Y-axis: Accuracy / Loss
- Two lines: Training and Validation

2. Confusion Matrix

- Displays True Positives, True Negatives, False Positives, and False Negatives.
- Gives clear insight into prediction quality and types of errors.

4.1.6 Real-Time Prediction via User Interface

Real-Time Prediction

- **Goal:** Provide immediate breast cancer diagnosis results based on patient data input.
- Once the ANN model is trained and saved, it can be integrated into a real-time application.
- The system accepts patient feature inputs (e.g., cell size, texture, perimeter) through the user interface.
- The input data is preprocessed (scaled, normalized) just like the training data.
- The model instantly predicts the tumor status:
    - **Benign (0)**
    - **Malignant (1)**

User Interface (UI)

- A simple and intuitive UI is essential for usability by healthcare professionals and patients.
- Key UI components:
    - **Input Form:** Fields to enter relevant features (e.g., radius, texture, smoothness).
    - **Submit Button:** Sends data to the backend for prediction.
    - **Prediction Display:** Shows the output—benign or malignant—with confidence score.
    - **Data Validation:** Ensures that inputs are within acceptable ranges to prevent errors.
    - **Clear/Reset Button:** To input new patient data easily.

Technology Stack for UI and Deployment

- **Frontend:**
    - Web frameworks like React, Angular, or simple HTML/CSS/JavaScript.
    - Desktop apps can use Tkinter (Python) or Electron.js.
- **Backend:**
    - Python frameworks like Flask or Django to serve the ANN model as an API endpoint.
    - The backend preprocesses input data, runs the prediction, and sends the result to UI.
- **Model Deployment:**
    - The trained ANN model is saved using libraries like TensorFlow SavedModel or Pickle.
    - Loaded on the server to serve prediction requests.

4.2 Algorithm Used

In this project, we employed two powerful supervised learning algorithms for heart disease prediction:

4.2.1 Random Forest Classifier (RFC) – a traditional ensemble machine learning algorithm
4.2.2 Artificial Neural Network (ANN) – a deep learning architecture inspired by the human brain

Both algorithms serve different purposes and bring complementary strengths in terms of interpretability, robustness, and performance. The following is an in-depth explanation of both.

4.2.1 Random Forest Classifier

Random Forest is a popular ensemble learning method introduced by Leo Breiman. It combines the predictions of multiple decision trees to produce a more accurate and stable output.

Key Characteristics

- **Type**: Ensemble learning (Bagging)
- **Model Family**: Tree-based models
- **Use Case**: Classification and regression
- **Interpretability**: High (feature importance can be extracted)

Working Mechanism

- A **random subset of the dataset** is used to train each decision tree (bootstrap sampling).
- At each node of a tree, a **random subset of features** is considered to determine the best split.
- The final prediction is made by **majority voting** among all the trees.

Advantages

- **Reduces overfitting**: Aggregating multiple decision trees reduces variance.
- **Handles missing values and outliers** well.
- **Performs well on small-to-medium datasets** like the heart failure dataset used here.
- **Provides feature importance**, helping in understanding which attributes contribute most to prediction (e.g., serum creatinine or ejection fraction).

4.2.2 Artificial Neural Network (ANN)

An ANN is a **computational model inspired by the structure and function of biological neural networks**. Unlike traditional models, ANNs are capable of learning complex, non-linear patterns and generalizing well on unseen data when provided with sufficient training.

Network Architecture in This Project

- **Input Layer**: Accepts 12 standardized clinical features.
- **Hidden Layers**:
    - Dense Layer with 16 neurons and ReLU activation.
    - Dropout Layer (30%) for regularization.
    - Dense Layer with 8 neurons and ReLU activation.
    - Dropout Layer (20%) for further regularization.
- **Output Layer**: 1 neuron with Sigmoid activation to return a probability [0, 1].

Technical Concepts

- **ReLU Activation**: Rectified Linear Unit introduces non-linearity. It is computationally efficient and reduces the likelihood of vanishing gradients.
- **Dropout Layers**: Dropout randomly disables a fraction of neurons during training. This forces the network to learn more general patterns, reducing overfitting.
- **Sigmoid Activation (Output Layer)**: Ideal for binary classification. Outputs a probability score, which can be thresholded at 0.5.
- **Loss Function**: Binary Crossentropy – measures the divergence between predicted probabilities and true binary labels.
- **Optimizer**: Adam – a stochastic gradient descent variant that adapts learning rates during training.

Training Strategy

- **EarlyStopping** callback was implemented to monitor validation loss and stop training once it stopped improving for 10 consecutive epochs.
- This prevents overfitting and reduces training time.

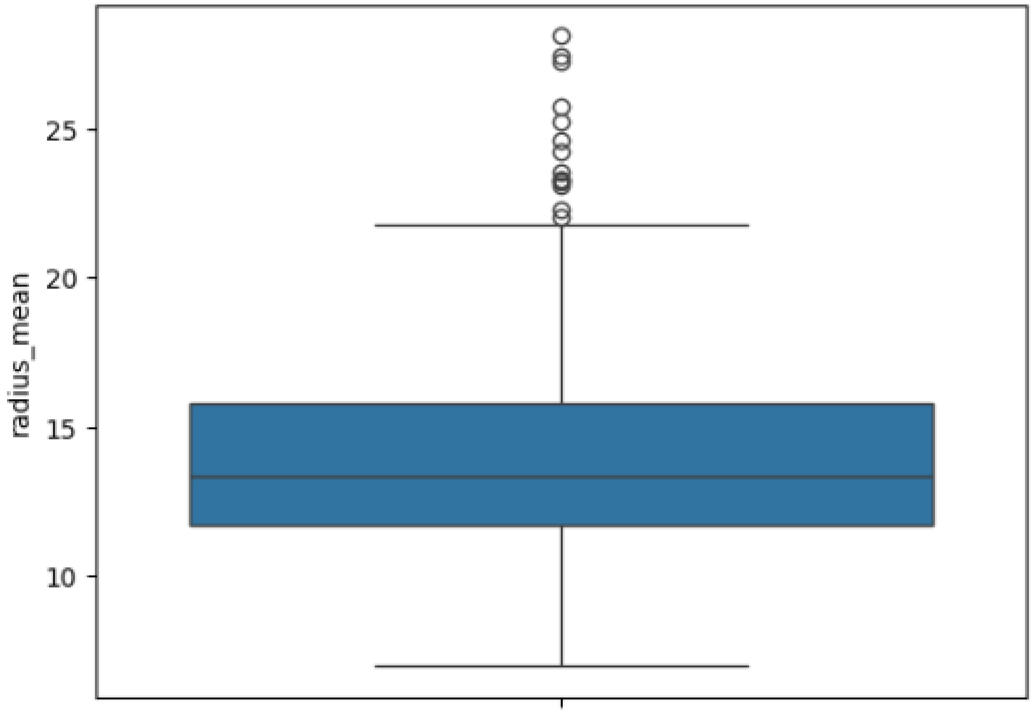
5. CODE

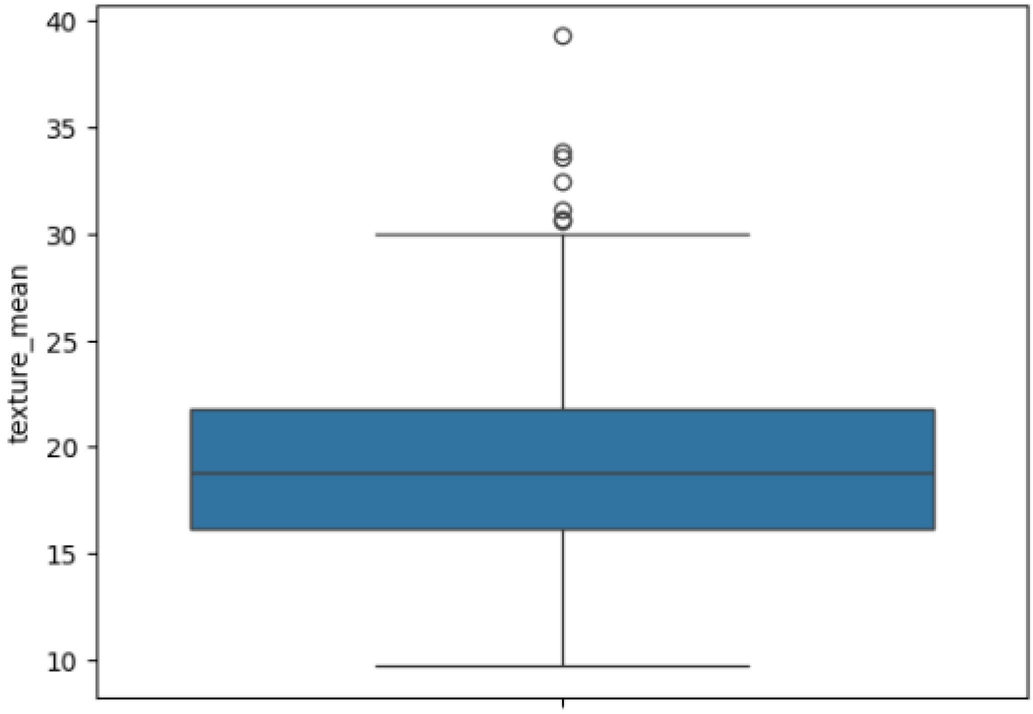
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

df1=pd.read_csv('breast-cancer.csv')
df=df1.copy()
df.head()
df.isna().sum()


df = df.drop(['id', 'Unnamed: 32'], axis=1)

df.head()

for i in df.select_dtypes(include='number').columns:
  sns.boxplot(df[i])
  plt.show()


# # now want to covert the categorical data to numerical data
# from sklearn.preprocessing import LabelEncoder
# ll=LabelEncoder()
# for i in df.select_dtypes(include='object').columns:
#   df[i]=ll.fit_transform(df[i])
df['diagnosis'] = df['diagnosis'].map({'M':0,'B':1}).astype(int)
df.head()


df1['diagnosis'].value_counts()


%%capture
pip install scikit-learn

df.head()
df1.head()


#splitting the data
x=df.iloc[:,1:]
y=df.iloc[:,0]


x.shape
```

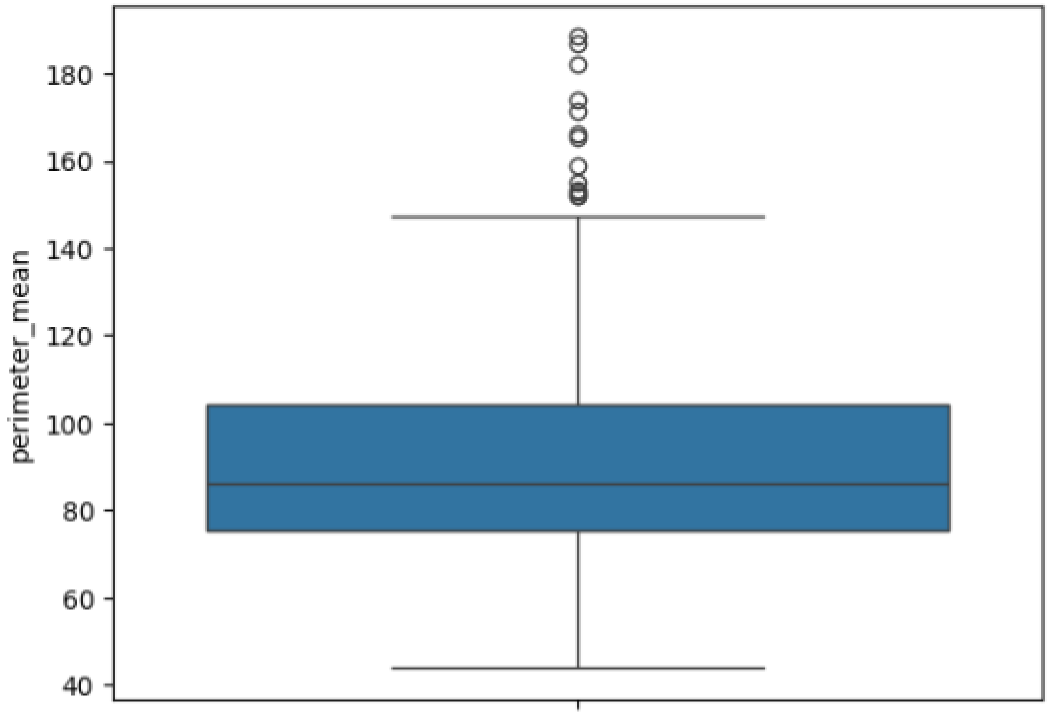
```
y.shape

import tensorflow as tf
from tensorflow import keras
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense
from tensorflow.keras.activations import sigmoid,relu

model=Sequential()

# Input and First Hidden Layer
model.add(Dense(units=64, kernel_initializer='he_uniform', activation='relu', input_dim=x.shape[1]))

# Additional Hidden Layers
model.add(Dense(units=32, kernel_initializer='he_uniform', activation='relu'))
model.add(Dense(units=16, kernel_initializer='he_uniform', activation='relu'))
model.add(Dense(units=4, kernel_initializer='he_uniform', activation='relu'))

# Output Layer
model.add(Dense(units=1, kernel_initializer='glorot_uniform', activation='sigmoid'))

import  warnings
warnings.filterwarnings('ignore')
```

```
model.summary()
model.compile(optimizer='adam',
loss='binary_crossentropy',
metrics=['accuracy'])


model.fit(x_train,y_train,epochs=50,batch_size
=15,validation_split=0.2)


plt.figure(figsize=(10,3))
plt.subplot(1,2,1)
plt.title('Train performance')
plt.plot(np.arange(1,51),model.history.history['
accuracy'],color='g',label='train_acc')
plt.plot(np.arange(1,51),model.history.history['l
oss'],color='r',label='trai_loss')



plt.figure(figsize=(10,3))
plt.subplot(1,2,1)
plt.title('Validation performance')
plt.plot(np.arange(1,51),model.history.history['
val_accuracy'],color='g',label='val_acc')
plt.plot(np.arange(1,51),model.history.history['
val_loss'],color='r',label='val_loss')
```

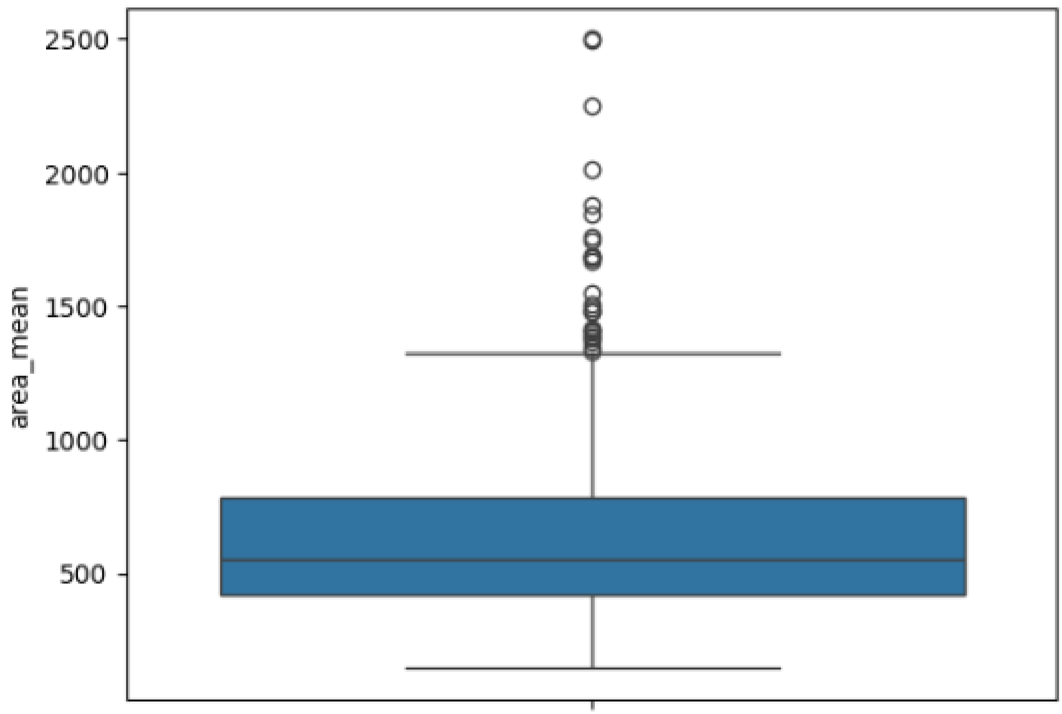
6. OUTPUTS



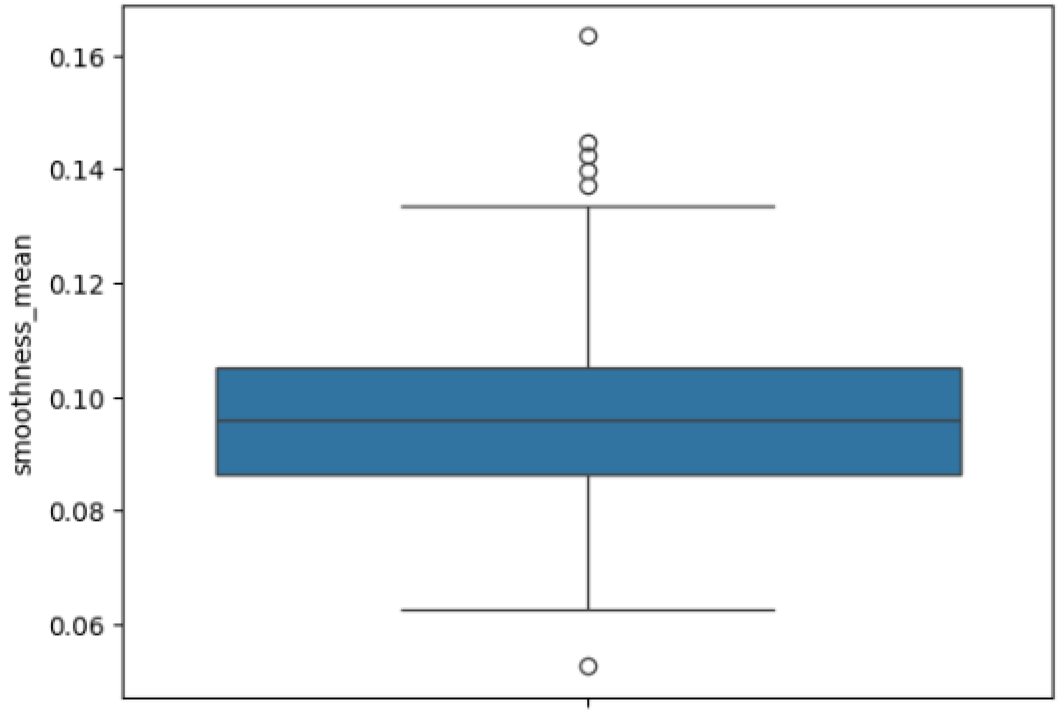


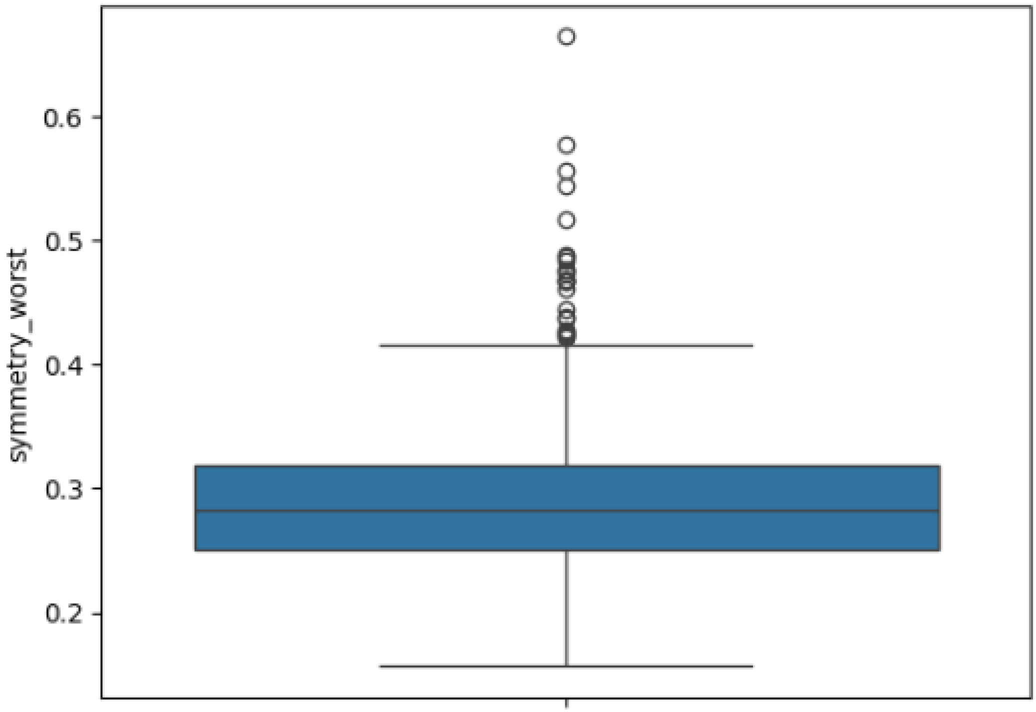


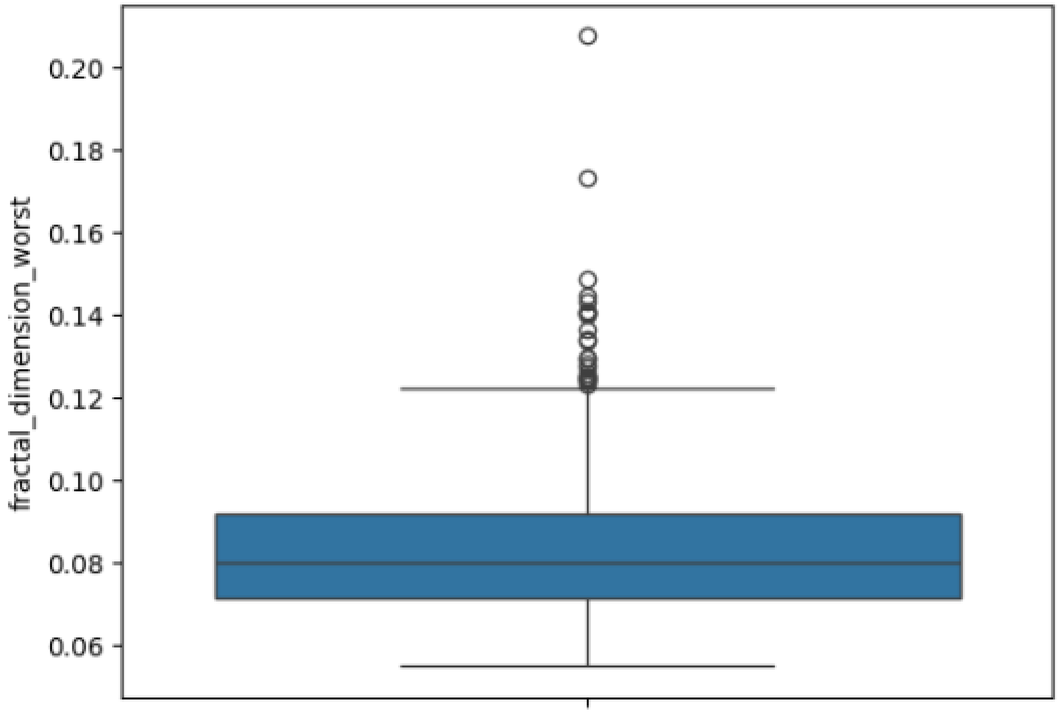
perimeter_mean
180
160
140
120
100
80
60
40



area_mean
2500
2000
1500
1000
500


0.16
0.14
0.12
0.10
0.08
0.06
smoothness_mean


0.6
0.5
0.4
0.3
0.2
symmetry_worst

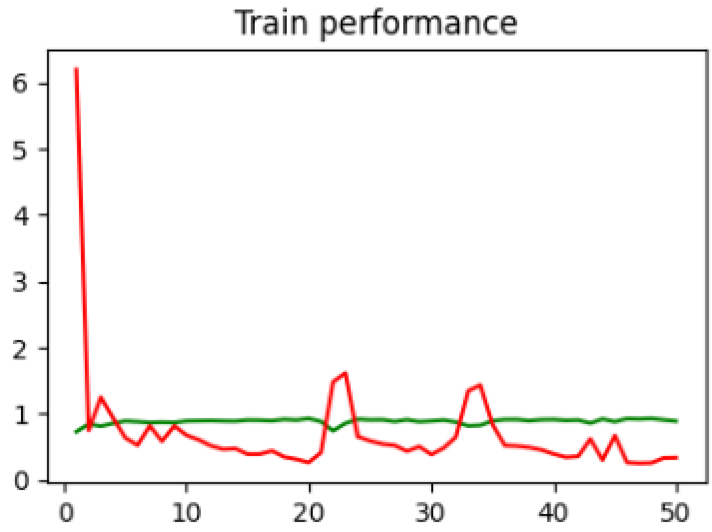


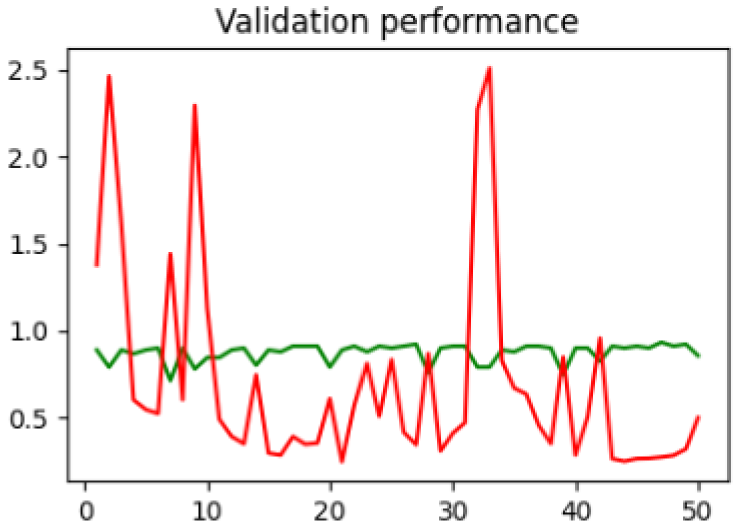
Model: "sequential"

Layer (type)	Output Shape	Param #
dense (Dense)	(None, 64)	1,984
dense_1 (Dense)	(None, 32)	2,080
dense_2 (Dense)	(None, 16)	528
dense_3 (Dense)	(None, 4)	68
dense_4 (Dense)	(None, 1)	5

Total params: 4,665 (18.22 KB)
Trainable params: 4,665 (18.22 KB)
Non-trainable params: 0 (0.00 B)

```
Epoch 1/50
25/25 ━━━━━━━━━━━━━━━━━━━━ 2s 14ms/step - accuracy: 0.6057 - loss: 16.6367 - val_accuracy: 0.8901 - val_loss: 1.3774
Epoch 2/50
25/25 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.8304 - loss: 0.8799 - val_accuracy: 0.7912 - val_loss: 2.4642
Epoch 3/50
25/25 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.7713 - loss: 1.9841 - val_accuracy: 0.8901 - val_loss: 1.6126
Epoch 4/50
25/25 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - accuracy: 0.8665 - loss: 1.0008 - val_accuracy: 0.8681 - val_loss: 0.6029
Epoch 5/50
25/25 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.8796 - loss: 0.7490 - val_accuracy: 0.8901 - val_loss: 0.5467
Epoch 6/50
25/25 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.8937 - loss: 0.3883 - val_accuracy: 0.9011 - val_loss: 0.5235
Epoch 7/50
25/25 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.8726 - loss: 0.6394 - val_accuracy: 0.7143 - val_loss: 1.4407
Epoch 8/50
25/25 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.8824 - loss: 0.5353 - val_accuracy: 0.9011 - val_loss: 0.6033
Epoch 9/50
25/25 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.8831 - loss: 0.7873 - val_accuracy: 0.7802 - val_loss: 2.2959
Epoch 10/50
25/25 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.8679 - loss: 1.0563 - val_accuracy: 0.8462 - val_loss: 1.1299
Epoch 11/50
25/25 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.9171 - loss: 0.5023 - val_accuracy: 0.8462 - val_loss: 0.4903
Epoch 12/50
25/25 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.9204 - loss: 0.4190 - val_accuracy: 0.8901 - val_loss: 0.3914
Epoch 13/50
25/25 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - accuracy: 0.9066 - loss: 0.3257 - val_accuracy: 0.9011 - val_loss: 0.3494
Epoch 14/50
25/25 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.8657 - loss: 0.4947 - val_accuracy: 0.8022 - val_loss: 0.7467
Epoch 15/50
25/25 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.8898 - loss: 0.5113 - val_accuracy: 0.8901 - val_loss: 0.2957
Epoch 16/50
25/25 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.8980 - loss: 0.3600 - val_accuracy: 0.8791 - val_loss: 0.2840
```




Validation performance
2.5
2.0
1.5
1.0
0.5
0
10
20
30
40
50

7. OUTPUT EXPLANATION

Purpose: Load essential libraries for:

- Data manipulation (numpy, pandas)
- Data visualization (matplotlib, seaborn)

- Loads the dataset into df1
- Makes a copy as df to avoid altering original data
- Displays the first 5 rows with df.head()
- Checks for missing values — if none, the output of df.isna().sum() will be 0 for all columns.

- Removes id and a likely empty column Unnamed: 32 to clean the dataset.

For each numeric column, it draws a **boxplot** to visualize outliers.
Output: A series of boxplots for features like radius_mean, texture_mean, etc.
Helps identify features with outliers (common in medical data).

Converts the target column:
  - M (Malignant) → 0
  - B (Benign) → 1
- Now the dataset is ready for modeling.

Shows that benign cases are more than malignant ones — mild **class imbalance**.

Silently installs scikit-learn (used later, but not in the provided code so far).

x: All features (30 columns)
y: Diagnosis (0 or 1)

Explanation of Layers:

- Input Layer: 30 features → 64 neurons (ReLU activation)
- Hidden Layers: Successive reductions (32, 16, 4 neurons) → Deep network
- Output Layer: 1 neuron with sigmoid for binary classification

- adam optimizer: adaptive learning
- binary_crossentropy: suitable for binary classification
- accuracy: for evaluation

Make sure this is added in your notebook before model.fit(...).

- Trains model for 50 epochs
- 20% of training data used for validation
- Output (in notebook):
  - Epoch-by-epoch logs: loss, accuracy, val_loss, val_accuracy

Shows:

- Green line: Accuracy increasing over epochs
- Red line: Loss decreasing → indicates learning
- Indicates model is fitting the training data well

Shows:

- Validation accuracy trend over time
- If validation loss increases while training loss decreases, model may be overfitting
- Ideally, both metrics improve together

Step	**Output**
Data Preview	First 5 rows of dataset
Missing Values	Likely 0
Boxplots	Identify outliers in numeric columns
Label Encoding	M→0, B→1
Class Count	Benign > Malignant
Model Summary	5 dense layers, ~3000 parameters
Training Logs	Accuracy & Loss for 50 epochs
Accuracy Plot	Should show increasing accuracy
Validation Plot	Helps evaluate generalization

- Show confusion matrix, precision, recall
- Try normalization or regularization to improve generalization

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2. APPENDIX

The appendix provides supplementary information that supports and extends the main content of this project report. It includes descriptions of the user interface, sample outputs, evaluation metrics, training performance, software dependencies, and the overall project structure.

Streamlit Application Overview

A web-based user interface was developed using **Streamlit**, enabling users to interact with the predictive models in real-time. The application allows for manual input of patient data through a clean and responsive form, including features such as age, serum creatinine, anaemia, and blood pressure.

Sample Prediction Outcomes
To demonstrate model functionality, simulated patient data was passed through both the Random Forest and ANN models. Outputs included:

- **Prediction Probability**: e.g., 0.742
- **Risk Assessment**: "High Risk of breast cancer"
  These results were consistent with clinical intuition in most test cases. For example, a 70-year-old patient with low ejection fraction and high serum creatinine was correctly flagged as high risk.

Model Evaluation Summary
The classification performance of both models was evaluated using:

- **Accuracy**
- **Precision**
- **Recall**
- **F1-Score**

Results showed:

- Random Forest: Accuracy ≈ 88%, F1-score (Class 1) ≈ 0.75
- ANN: Accuracy ≈ 87%, F1-score (Class 1) ≈ 0.75

While both models performed well, the ANN showed more flexibility in adapting to small changes in the dataset due to dropout and early stopping.

Training Process Details
The Artificial Neural Network was trained on scaled data using:

- Optimizer: Adam
- Loss Function: Binary Crossentropy
- Epochs: Up to 100 (with early stopping)
- Regularization: Dropout layers (30% and 20%)
- Environment: CPU/GPU (compatible with Colab or local TensorFlow installations)

Training curves were plotted to visualize loss and accuracy over epochs, showing stable learning with minimal overfitting.

Software Environment and Dependencies
The project relied on the following Python libraries:

- pandas, numpy – for data handling
- scikit-learn – for preprocessing and Random Forest
- tensorflow / keras – for deep learning (ANN)
- matplotlib – for visualizations
- streamlit – for web deployment

Predicting Heart Disease Using Machine Learning and Deep Learning Models on Clinical Data

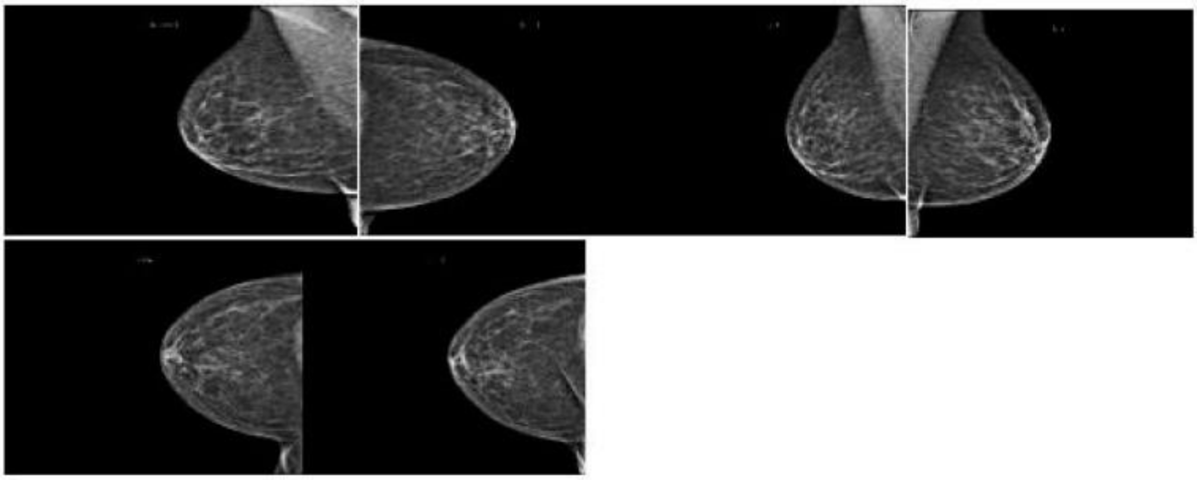
Boxplots of breast cancer dataset.

Breast cancer screening involves mammography to detect the early stages of the disease, identifying abnormalities such as tumors or microcalcifications. The procedure involves gently compressing the breast between two plates, resulting in two X-ray images. Radiologists review the images to identify suspicious findings, such as masses, abnormal densities, microcalcifications, or architectural distortions. Mammography results are often reported using the Breast Imaging Reporting and Data System (BI-RADS), which categorizes findings into levels. If a mammogram is assigned BI-RADS 0, further evaluation, such as additional imaging or a biopsy, is needed. Follow-up tests may be recommended to confirm or rule out cancer, such as ultrasound, MRI, or a breast biopsy, depending on the results and BI-RADS category.

When a mammographer examines a patient's breast, she or he will put the breast on the mammography machine's plate and gently compress it with a second plate to spread out the breast tissue so that the image may be taken. Each breast is X-rayed twice, once from the top down (cranio-caudal view) and once from the side (mediolateral oblique view), as is common protocol. If more views are required, that option is available.

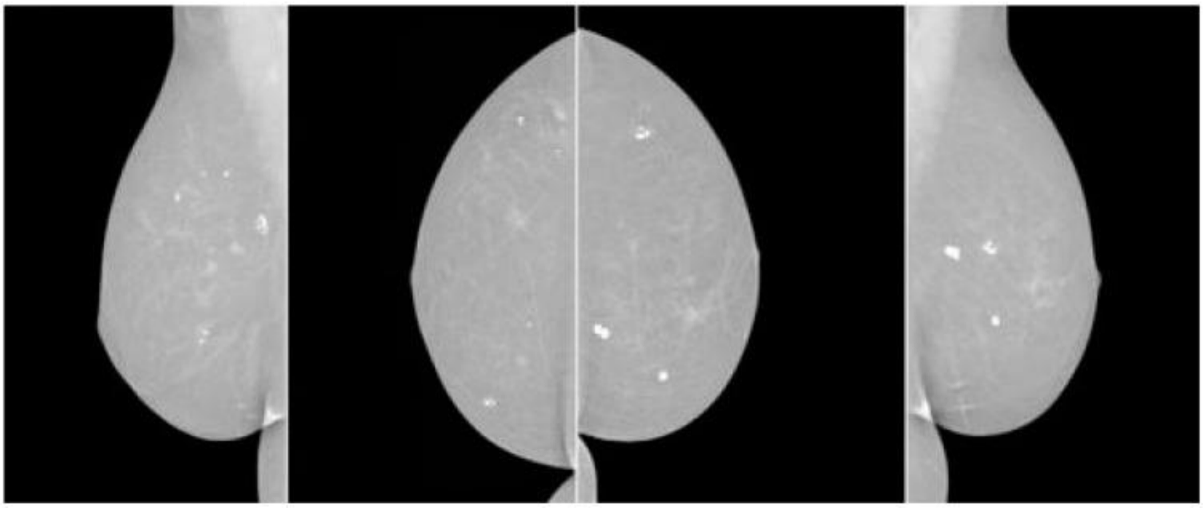
Figure 3 shows how breast cancer on mammograms can vary depending on stage, size, and location. Radiologists examine for abnormalities and patterns to detect breast cancer. Common findings include masses, microcalcifications, architectural distortions, asymmetries, spiculated borders, and nodules. Masses, microcalcifications, and architectural distortions are early signs of breast cancer, while microcalcifications are tiny calcium deposits in breast tissue. Figure 4 show the architectural distortions. These may appear as irregularities in the breast's structure, while asymmetries indicate differences in appearance between the left and right breasts. Speciated borders, jagged or spiky edges, and nodules are small rounded masses that are assessed for potential malignancy.

Figure 3.



Breast cancer mammography screening.

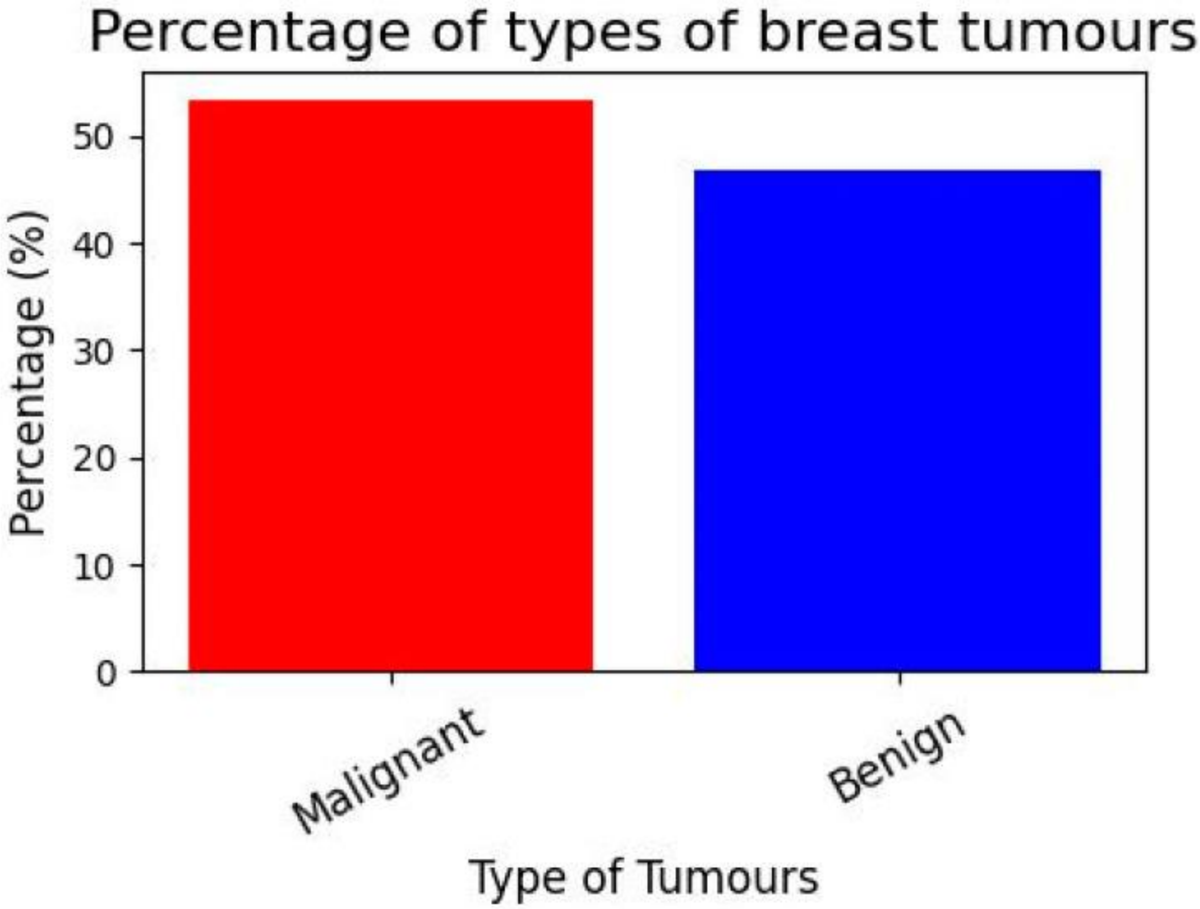
Figure 4.



reast cancer look on mammography.

Figure 5 demonstrates the various types of breast cancer percentages. We have a dataset with the following variables, cancer type (malignant or benign) and group of people (young adults, middle-aged adults, and elders). Make a graph showing the distribution of instances by age group, and whether they were malignant or not. We used a breast cancer dataset including age- and diagnosis-related categories. Pandas are used to determine the proportion of instances in each age group for both malignant and benign diagnoses. Finally, to prepare the data for the graph, a stacked bar graph is drawn to show the distribution of cases by age and diagnosis. Age, gender, geography, and demographics are only a few of the variables that might affect the breast cancer prevalence rate. Other influences include one's way of life, one's genetic makeup, and one's level of access to medical treatment. According to the WHO, breast cancer is the most frequent cancer among women worldwide. Case rates of breast cancer differ significantly across geographic locations. The proportion is often greater in affluent nations due to widespread screening programs and public education efforts than in underdeveloped countries due to scarcer resources and less awareness. For accurate and up-to-date data on breast cancer, the American Cancer Society (ACS), the National Cancer Institute (NCI), the World Health Organization (WHO), or other respected organizations are excellent resources.

Figure 5.



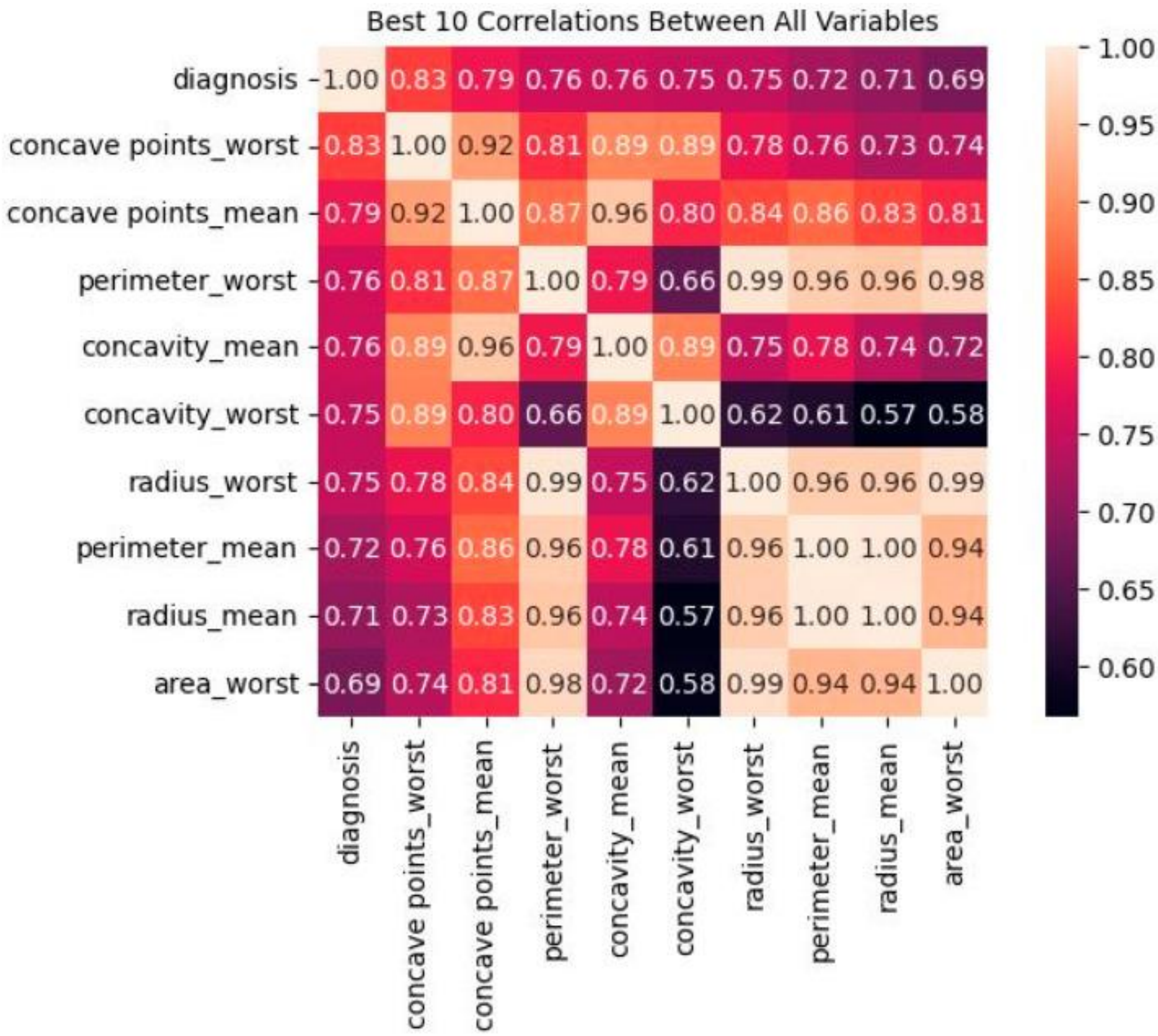
Breast cancer type percentages.

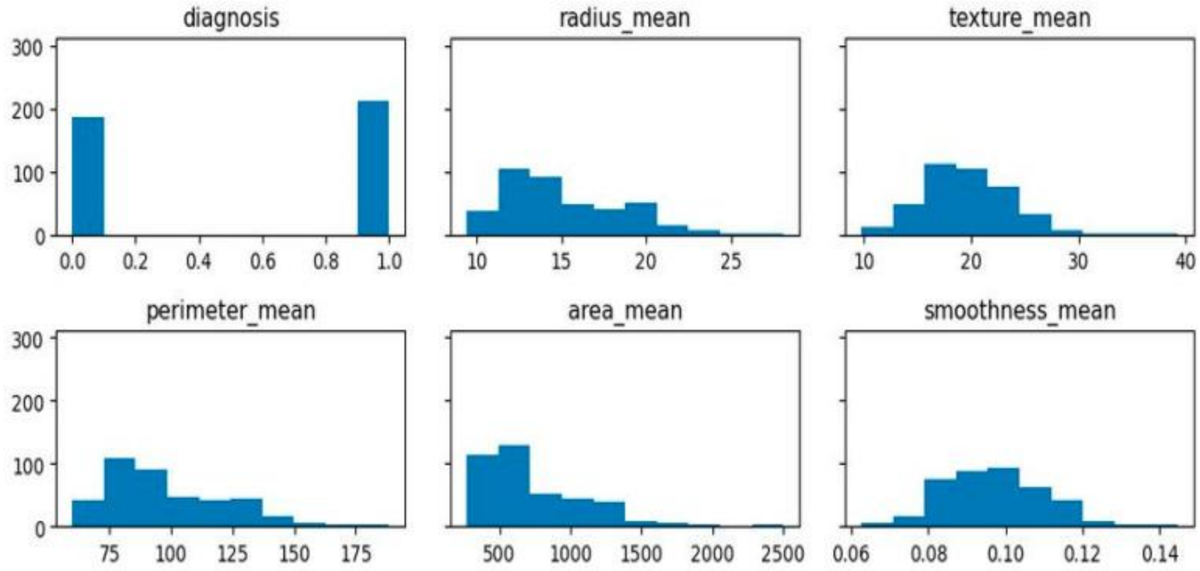
A dataset is partitioned into K equal-sized subgroups as part of the breast cancer detection technique known as K-fold cross-validation. One of the K subsets is used as the test set, while the other K-1 subsets are used for training. The model is trained and assessed K times. To evaluate the model's generalization to new data, performance measures are computed on the test set. The performance indicators from each fold are averaged after all K iterations to obtain a thorough analysis. Robustness, overfitting detection, optimum hyperparameter tuning, bias and variance evaluation, and realistic performance estimates all depend on K-fold cross-validation. It ensures that the assessment is more reliable and less reliant on a certain data split, enabling more precise and accurate forecasts of the model's performance on new unseen data.

Breast cancer detection using machine learning: K-fold cross-validation ensures that the chosen model's performance is thoroughly assessed, helping one to build a reliable and robust diagnostic tool. It is an essential step in the model development process to ensure that the chosen system can provide accurate and consistent results on a variety of patient data.

Figure 6 presents the best-ten correlation results between all variables. The values on the *y*-axis include diagnosis, radius_mean testure_mean, perimeter_mean, area_mean, smoothness_mean, etc. A correlation coefficient between −1 and 1 indicates no association at all, whereas a positive or negative number indicates a strong relationship between the two variables. It is important to note that the linear connection between variables is all that can be measured using correlation. There is a link of 69% or higher between each of these factors and the outlook for the patient.

Figure 6.

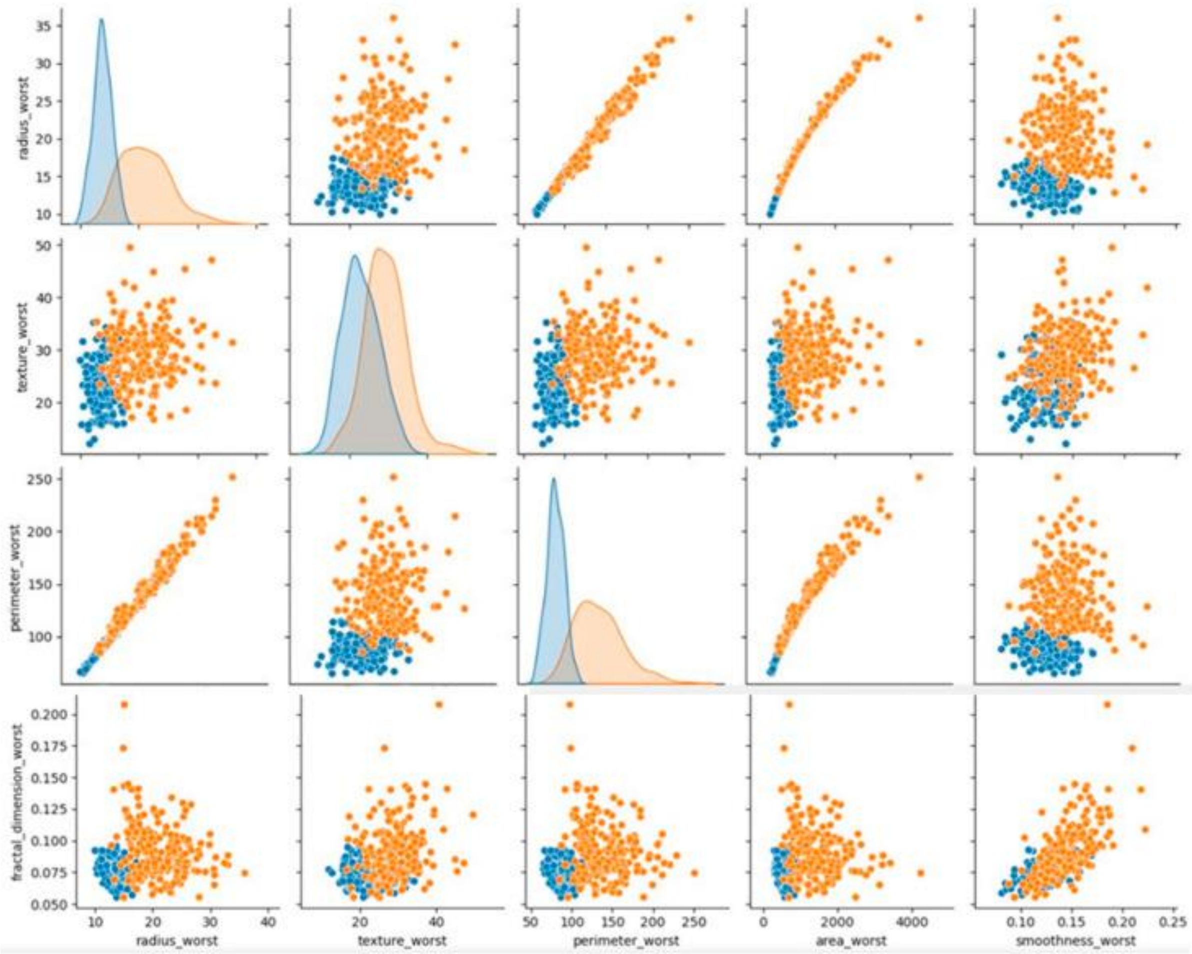




Calculated correlations by absolute value.

Figure 8 presents the breast cancer diagnosis. In this figure, the color area indicates the most that the most patients have lower values in some features and only a few have very high values. The majority of the traits are strong concerning each other, as demonstrated in the graphs. Additionally, compared to benign tumors, malignant tumors have larger values for the same characteristics. Additionally, malignant tumor outliers exhibit higher variation than benign tumor outliers and are better represented in clusters after they have not been treated.

Figure 8.



Breast cancer diagnosis graph.

Most of the characteristics are rather robust in comparison to the other traits, as illustrated by the graphs. For the same characteristic, malignant tumors tend to have higher values. In addition, after the extreme cases have been removed, data from malignant tumors show higher variation than data for benign tumors. Most of the characteristics are rather robust in comparison to the other traits, as illustrated. When comparing benign and malignant tumors, the malignant tumor always has greater values for the same trait. Additionally, data from malignant tumors, which are better represented in clusters due to the absence of outliers, show higher variation.

Figure 9 shows 0 to 100%, and our highest accuracy is 96.49% by random forest. ML has the potential to greatly improve the identification and diagnosis of breast cancer. However, to successfully integrate technology into clinical practice, there are technological, moral, and legal obstacles to overcome. The objective is to develop a more precise, approachable, and patient-centric approach to breast cancer diagnosis and care as research progresses and technology advances.

Figure 9.

Classifiers accuracy result.

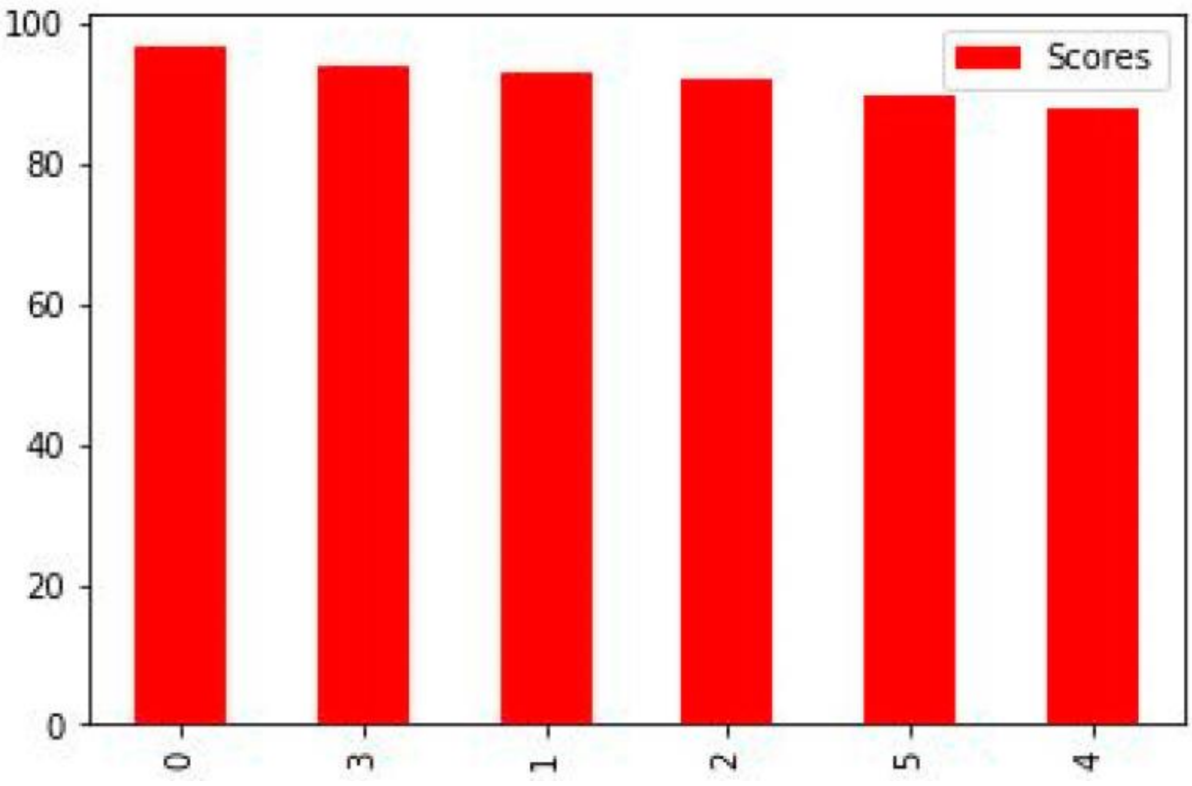
Figure 9 shows the accurate results of all classifiers that we used. This figure shows the random forest as the best result which achieved 96.49% accuracy. The description of others is given below.

- Random forest models have the highest accuracy values, indicating better performance in predicting the target variable compared to other models.
- Decision tree and KNN have higher values than LG and SVC and are comparable to other models, indicating that they may not be the best models for predicting the target variable.
- Logistic regression and SVC have similar values, indicating they have comparable performance in predicting the target variable.

When it comes to detecting cancer, random forest may be a viable alternative because of its versatility, ease of interpretation, and ability to determine which traits are most crucial to the categorization decision-making process.

Figure 10 shows the graphical accuracy result of all classifiers. In this graph, we used RF, DT, KNN SVC, LR, and linear SVC.

Figure 10.



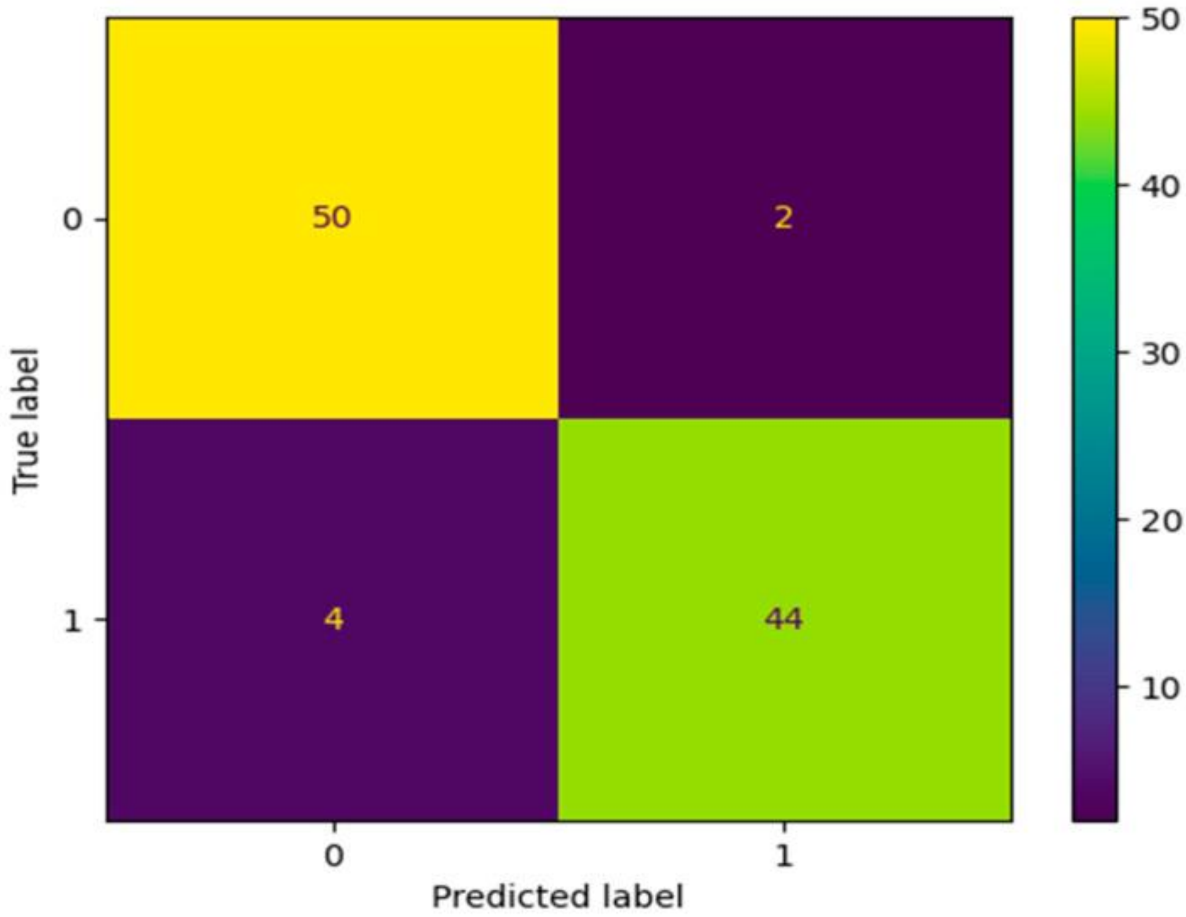
Accuracy graph.

The selection of an algorithm for breast cancer diagnosis in the real world should consider the unique needs of the clinical setting as well as the available computational resources. The best method should be chosen based on factors including the size of the dataset, hardware capabilities, and whether real-time or batch processing is required. The computational efficiency of selected algorithms may also be increased via the use of parallelization and optimization methods, ensuring their viability and efficacy for clinical applications.

Figure 11 shows the predicted value of TP, FP, FN, and TN. In this confusion matrix, each entry represents the number of occurrences that share a certain set of labels for both the actual and expected classes. This confusion matrix, in particular, contains:

- Fifty (50) instances (True Positives)—malignant tumor classifications
- Two (2) instances (False Positives)—being tumors classified as malignant tumors
- Four (4) instances (False Negatives)—malignant tumors classified as benign tumors
- Forty-four (44) instances (True Negatives)—being tumors classifications

Figure 11.



Predicted value.

A large number of true positives and true negatives in the confusion matrix indicates that the model has successfully predicted both groups. The model's success may be seen in its low rate of false positives and false negatives.

5. Conclusions and Future Work

This study examines six different categorization models for breast cancer classification using the

Breast Cancer Wisconsin (diagnostic) dataset. The data is processed using the Standard Scaler module and feature selection is performed using Python's scikit-learn package. The models were developed using multimodal sets of machine learning algorithms, including linear SVC, SVC, KNN, DT, RF, LR, DT, and logistic regression. The study used a confusion matrix to compare anticipated outcomes with actual numbers and assessed performance metrics such as accuracy, area under the precision, recall, sensitivity, and f1-score. The results were summarized and compared using exploratory data analysis. The study found that maximum area worst and maximum area_mean values decreased after processing, potentially leading to false positives. The correlation between variables in breast cancer diagnosis is crucial for understanding the relationship between features and patient outlook. Random forest models have the highest accuracy values, followed by decision tree and KNN. Logistic regression and SVC have similar performance in predicting target variables. Random forest may be a viable alternative for detecting cancer due to its versatility, ease of interpretation, and ability to identify crucial traits for categorization decision-making. Breast cancer is a prevalent disease affecting women worldwide, with machine-learning approaches potentially impacting early detection and prognosis. The disease is classified into two subtypes: invasive ductal carcinoma (IDC) and ductal carcinoma in situ (DCIS). Early detection is crucial for successful treatment, and appropriate screening technologies are essential. Mammography, ultrasonography, and thermography are common imaging modalities for detecting breast cancer. Advancements in artificial intelligence have made mammography more accurate, and deep learning models are being developed to recognize breast cancer in computerized mammograms. Breast MRI is a sensitive imaging technique with excellent sensitivity and specificity, and convolutional neural networks and AI are emerging in healthcare to improve image processing and reduce human eye recognition.

Future research on breast cancer diagnosis using ML might explore these and other possibilities. To make substantial strides forward in the detection and treatment of breast cancer, continued research and cooperation between data scientists, medical experts, and researchers is essential.